Global Weather Trend Forecasting
PM Accelerator Technical Assessment

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Project Overview

Introduction:

Weather forecasting plays an important role in understanding climate conditions and supporting decision-making in agriculture, transportation, disaster management, and environmental planning. Accurate weather prediction helps governments, organizations, and individuals prepare for changing weather conditions and reduce the impact of extreme weather events.

The objective of this project is to analyze the **Global Weather Repository** dataset and build a complete weather forecasting pipeline using Python and machine learning techniques. The project begins with data preprocessing, including handling missing values, removing duplicate records, detecting and removing outliers, and normalizing numerical features. After preparing the dataset, exploratory data analysis is performed to understand weather patterns, relationships between variables, and climate variations across different countries.

Advanced analytical techniques such as anomaly detection are used to identify unusual weather observations. Multiple machine learning models, including Linear Regression, Random Forest, XGBoost, LightGBM, and an Ensemble model, are developed and evaluated to forecast weather conditions. Model performance is compared using standard evaluation metrics to determine the most effective forecasting approach.

This project demonstrates an end-to-end data science workflow, covering data cleaning, visualization, statistical analysis, predictive modeling, and performance evaluation. The results provide meaningful insights into global weather trends while illustrating practical applications of machine learning in environmental data analysis.

2. Project Objectives

The primary objective of this project is to analyze global weather data and develop a machine learning model capable of forecasting temperature trends. The project follows a complete data science workflow, beginning with data preprocessing and ending with model evaluation.

The specific objectives are:

- Clean and preprocess the weather dataset.
- Handle missing values and duplicate records.
- Detect and remove outliers using the IQR method.
- Normalize numerical features to improve model performance.
- Perform Exploratory Data Analysis (EDA) to understand weather patterns.
- Analyze relationships between weather variables using correlation analysis.
- Detect unusual weather observations using the Isolation Forest algorithm.
- Build multiple machine learning models for temperature forecasting.
- Compare model performance using evaluation metrics.
- Identify the most accurate forecasting model.

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3. Dataset Description

This project uses the **Global Weather Repository** dataset, which contains weather observations collected from multiple countries around the world. The dataset includes atmospheric measurements, geographical information, and environmental indicators that are useful for weather analysis and forecasting.

The dataset provides a comprehensive view of global weather conditions and is suitable for both exploratory analysis and predictive modeling.

Important Features Used

Feature	**Description**
temperature_celsius	Air temperature in degrees Celsius
humidity	Relative humidity percentage
pressure_mb	Atmospheric pressure (millibars)
wind_kph	Wind speed (km/h)
precip_mm	Precipitation (millimeters)
cloud	Cloud cover percentage
visibility_km	Visibility (kilometers)
uv	UV index
air_quality	Air quality measurements
latitude	Geographic latitude
longitude	Geographic longitude
country	Country name
last_updated	Date and time of weather observation

```
df = pd.read_csv("GlobalWeatherRepository.csv")
df.head()
```

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	conditio
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	5/16/24 13:15	26.6	79.8	Partl
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1715849100	5/16/24 10:45	19.0	66.2	Partl
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	5/16/24 9:45	23.0	73.4	
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1715849100	5/16/24 10:45	6.3	43.3	Ligl
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	5/16/24 9:45	26.0	78.8	Partl

5 rows × 41 columns

Figure 1: First five rows of the dataset.

```
df.info()

df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 118448 entries, 0 to 153580
Data columns (total 41 columns):
 #   Column                        Non-Null Count   Dtype
---  ------                        --------------   -----
 0   country                       118448 non-null  object
 1   location_name                 118448 non-null  object
 2   latitude                      118448 non-null  float64
 3   longitude                     118448 non-null  float64
 4   timezone                      118448 non-null  object
 5   last_updated_epoch            118448 non-null  int64
 6   last_updated                  118448 non-null  datetime64[ns]
 7   temperature_celsius           118448 non-null  float64
 8   temperature_fahrenheit        118448 non-null  float64
 9   condition_text                118448 non-null  object
 10  wind_mph                      118448 non-null  float64
 11  wind_kph                      118448 non-null  float64
 12  wind_degree                   118448 non-null  int64
 13  wind_direction                118448 non-null  object
 14  pressure_mb                   118448 non-null  int64
 15  pressure_in                   118448 non-null  float64
 16  precip_mm                     118448 non-null  float64
 17  precip_in                     118448 non-null  float64
 18  humidity                      118448 non-null  float64
 19  cloud                         118448 non-null  int64
 20  feels_like_celsius            118448 non-null  float64
 21  feels_like_fahrenheit         118448 non-null  float64
 22  visibility_km                 118448 non-null  float64
 23  visibility_miles              118448 non-null  int64
 24  uv_index                      118448 non-null  float64
 25  gust_mph                      118448 non-null  float64
 26  gust_kph                      118448 non-null  float64
 27  air_quality_Carbon_Monoxide   118448 non-null  float64
 28  air_quality_Ozone             118448 non-null  float64
 29  air_quality_Nitrogen_dioxide  118448 non-null  float64
 30  air_quality_Sulphur_dioxide   118448 non-null  float64
 31  air_quality_PM2.5             118448 non-null  float64
 32  air_quality_PM10              118448 non-null  float64
 33  air_quality_us-epa-index      118448 non-null  int64
 34  air_quality_gb-defra-index    118448 non-null  int64
 35  sunrise                       118448 non-null  object
 36  sunset                        118448 non-null  object
 37  moonrise                      118448 non-null  object
 38  moonset                       118448 non-null  object
 39  moon_phase                    118448 non-null  object
 40  moon_illumination             118448 non-null  int64
dtypes: datetime64[ns](1), float64(22), int64(8), object(10)
```

Figure 2 &3 : Dataset information showing columns and data types. Statistical summary of the numerical features.

4.Explanation of Dataset:

The dataset contains weather observations from different countries, allowing the analysis of weather patterns across various geographic regions. The numerical variables include temperature, humidity, wind speed, precipitation, pressure, visibility, and air quality measurements, while categorical variables such as country provide geographical context. The last_updated column was converted into a datetime format to support time-series analysis and forecasting. Before building machine learning models, the dataset underwent preprocessing to improve data quality and ensure reliable prediction results.

5.Data Preprocessing

Before analyzing the data, the dataset was cleaned to improve its quality. Cleaning the data helps remove errors and prepares it for machine learning. The preprocessing steps included checking missing values, removing duplicate records, converting the date column, removing outliers, and normalizing numerical features.

5.1 Missing Values

The dataset was checked for missing values using the isnull().sum() function. Missing values can reduce the accuracy of analysis, so they were identified before moving to the next step.

Data Cleaning

Missing Values

```
print(df.isnull().sum())
```

```
country                         0
location_name                   0
latitude                        0
longitude                       0
timezone                        0
last_updated_epoch              0
last_updated                    0
temperature_celsius             0
temperature_fahrenheit          0
condition_text                  0
wind_mph                        0
wind_kph                        0
wind_degree                     0
wind_direction                  0
pressure_mb                     0
pressure_in                     0
precip_mm                       0
precip_in                       0
humidity                        0
cloud                           0
feels_like_celsius              0
feels_like_fahrenheit           0
visibility_km                   0
visibility_miles                0
uv_index                        0
gust_mph                        0
gust_kph                        0
air_quality_Carbon_Monoxide     0
air_quality_Ozone               0
air_quality_Nitrogen_dioxide    0
air_quality_Sulphur_dioxide     0
air_quality_PM2.5               0
air_quality_PM10                0
air_quality_us-epa-index        0
air_quality_gb-defra-index      0
sunrise                         0
sunset                          0
moonrise                        0
moonset                         0
moon_phase                      0
moon_illumination               0
dtype: int64
```

Explanation:

The output shows the number of missing values in each feature. The missing values were handled so that the dataset could be used for analysis and model training.

5.2 Duplicate Records

Duplicate rows were removed to ensure that each weather observation appears only once in the dataset.

Remove Duplicates

```
df.drop_duplicates(inplace=True)

print(df.shape)
```

```
(153581, 41)
```

Explanation:

Removing duplicate records improves data quality and prevents repeated information from affecting the analysis.

5.3 Outlier Detection and Removal

Outliers are unusually high or low values that can reduce the accuracy of machine learning models. The Interquartile Range (IQR) method was used to detect and remove these values.

Remove Outliers

```
columns = [
    "temperature_celsius",
    "humidity",
    "wind_kph",
    "precip_mm"
]

Q1 = df[columns].quantile(0.25)
Q3 = df[columns].quantile(0.75)

IQR = Q3 - Q1

df = df[
    ~((df[columns] < (Q1 - 1.5 * IQR)) |
      (df[columns] > (Q3 + 1.5 * IQR))).any(axis=1)
]

print(df.shape)
```

```
... (118448, 41)
```

Explanation:

Removing outliers makes the data more balanced and helps improve the performance of machine learning models.

5.4 Data Normalization

The numerical features were normalized using **StandardScaler**. Normalization scales the numerical values to a similar range, allowing machine learning algorithms to learn more effectively.

Normalize

```
scaler = StandardScaler()

df[columns] = scaler.fit_transform(df[columns])

df.head()
```

...

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	condition_text
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	2024-05-16 13:15:00	0.548668	79.8	Partly Cloudy
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	2024-05-16 09:45:00	0.145022	73.4	Sunny
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	2024-05-16 09:45:00	0.481393	78.8	Partly cloudy
5	Antigua and Barbuda	Saint John's	17.12	-61.85	America/Antigua	1715849100	2024-05-16 04:45:00	0.481393	78.8	Partly cloudy
6	Argentina	Buenos Aires	-34.59	-58.67	America/Argentina/Buenos_Aires	1715849100	2024-05-16 05:45:00	-1.536834	46.4	Clear

5 rows × 41 columns

Explanation:

After normalization, the numerical features are on a similar scale, which helps improve model training and prediction accuracy.

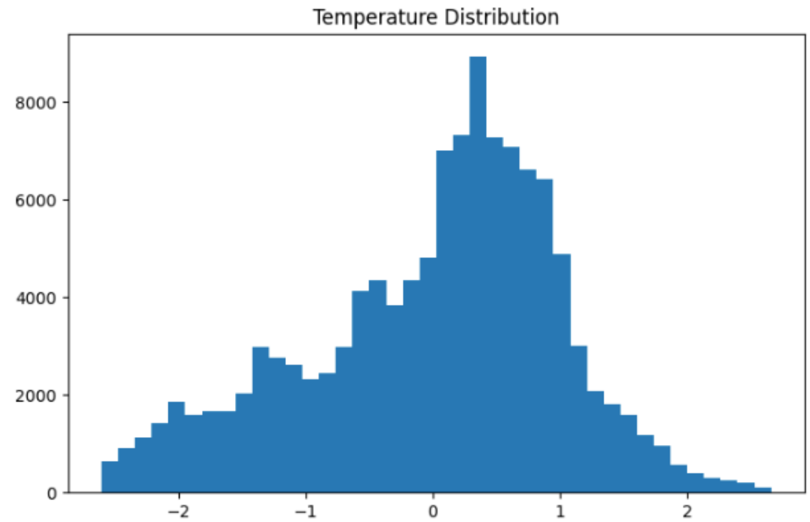
6.Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to better understand the weather dataset before building machine learning models. Different charts were used to identify patterns, trends, and relationships between weather variables.

6.1 Temperature Distribution

A histogram was created to show how temperature values are distributed in the dataset.

...

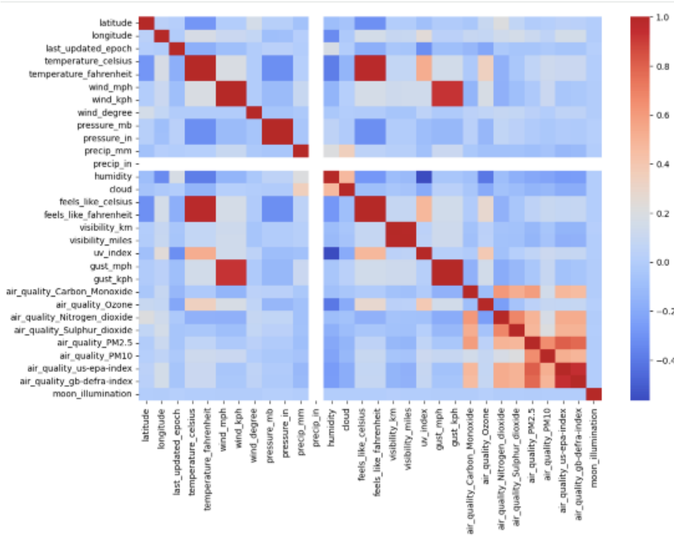


Explanation:

The histogram shows that most temperature values are concentrated within a specific range, while very high and very low temperatures occur less frequently.

6.2 Correlation Heatmap

A correlation heatmap was created to understand the relationship between numerical weather features.

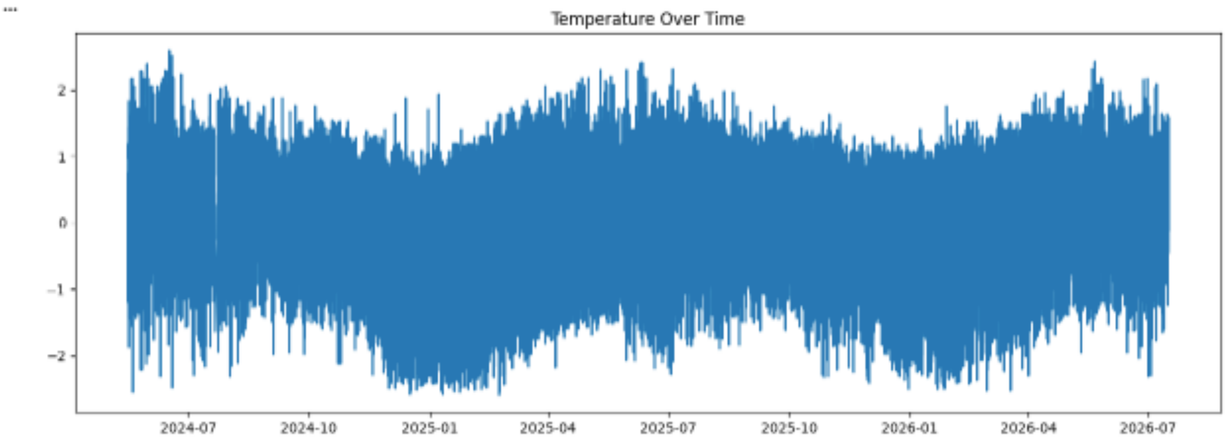


Explanation:

The heatmap helps identify which weather variables are strongly related. Features with higher correlations can be useful for building prediction models.

6.3 Temperature Trend Over Time

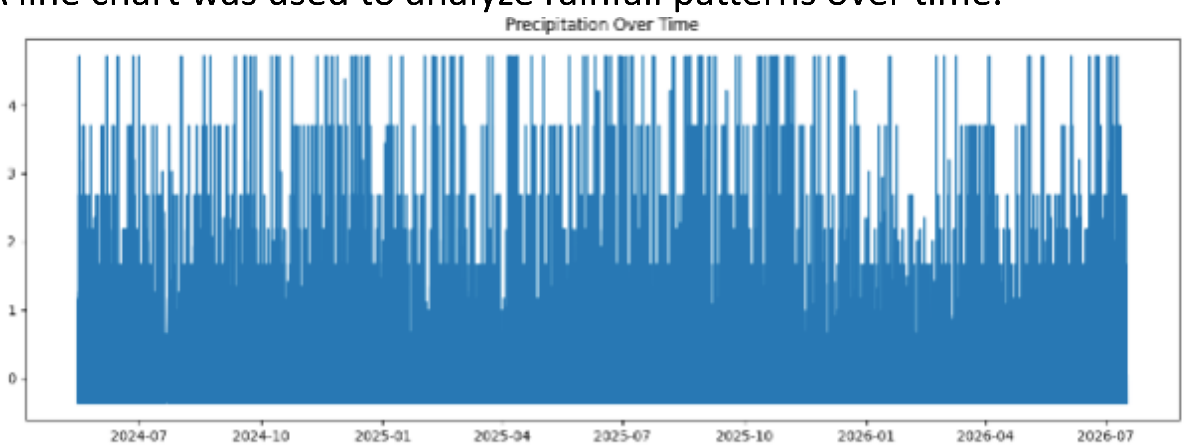
A line chart was created to observe changes in temperature over time.



The graph shows how temperature changes across different dates, helping identify seasonal patterns and fluctuations.

6.4 Precipitation Trend Over Time

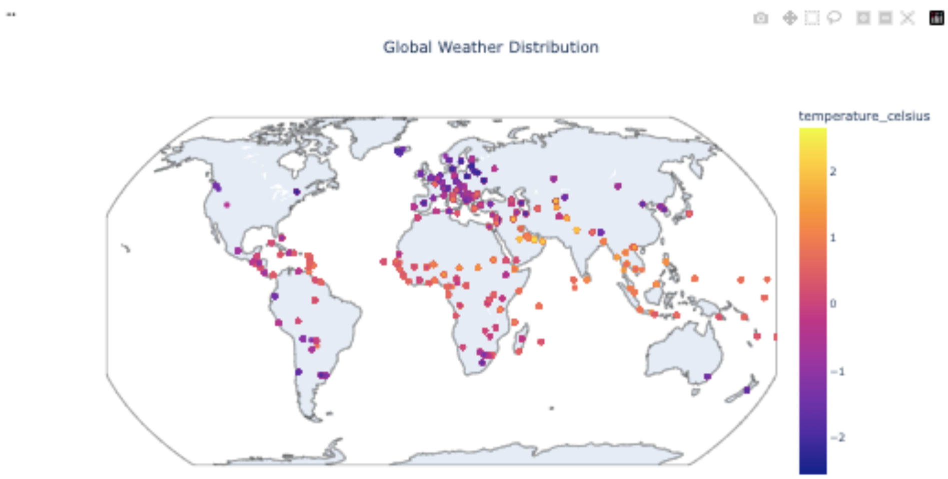
A line chart was used to analyze rainfall patterns over time.



The graph shows changes in rainfall over time and helps identify periods with higher or lower precipitation.

6.5 Global Weather Distribution

A world map was created to display weather observations from different countries.



Explanation:

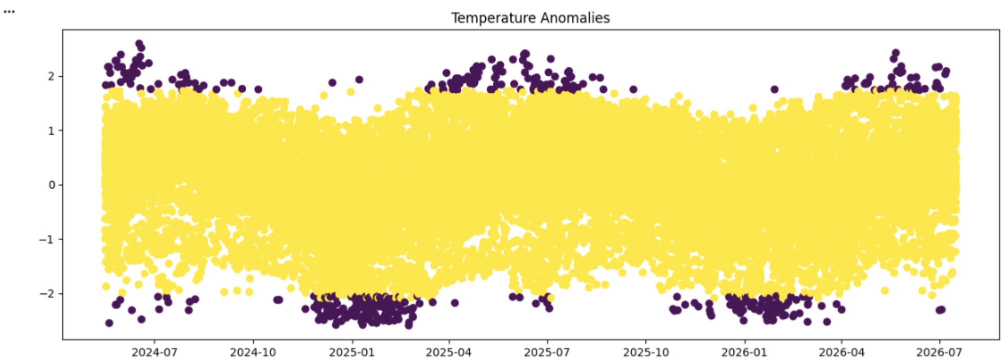
The map displays weather data collected from different parts of the world. The colors represent temperature values, making it easy to compare weather conditions across locations.

7. Anomaly Detection

Anomaly detection was performed to identify unusual weather observations in the dataset. These unusual observations, also called **outliers or anomalies**, may represent extreme weather conditions or incorrect data entries. Detecting them helps improve data quality and provides additional insights into the dataset.

Isolation Forest

Isolation Forest is an unsupervised machine learning algorithm that identifies anomalies by isolating observations that are different from the majority of the data.



Explanation:

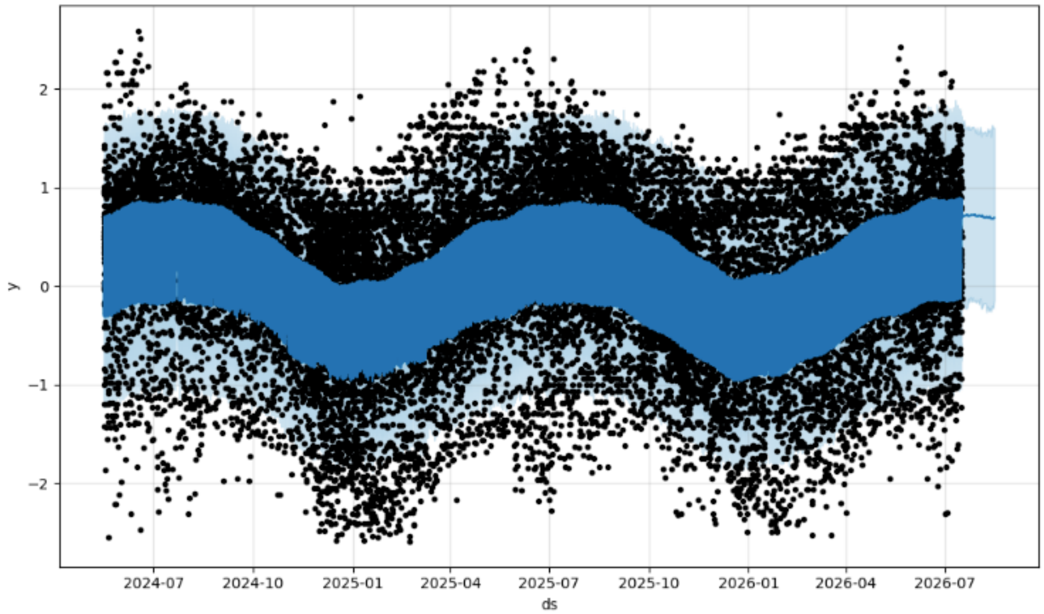
The algorithm classified each observation as either **normal** or **anomalous**. Normal observations represent typical weather conditions, while anomalous observations indicate unusual weather patterns that may require further investigation.

8.Feature Importance

Feature importance analysis was performed to identify which weather variables had the greatest impact on predicting temperature. Understanding feature importance helps explain how the machine learning model makes predictions and highlights the most influential variables.

8.1 Feature Importance Analysis

The feature importance scores were calculated using the trained machine learning model. Features with higher scores have a greater influence on the prediction of temperature.



9.Machine Learning Models

After completing data preprocessing, exploratory data analysis, and anomaly detection, several machine learning models were trained to predict temperature values. The performance of each model was evaluated using **Mean Absolute Error (MAE)**. A lower MAE indicates better prediction accuracy.

Model Evaluation

The performance of the machine learning models was evaluated using **Mean Absolute Error (MAE)**. MAE measures the average difference between the predicted and actual temperature values. A lower MAE indicates better prediction accuracy.

```python
print("Random Forest MAE:", mean_absolute_error(y_test,pred))

print("XGBoost MAE:", mean_absolute_error(y_test,pred_xgb))

print("LightGBM MAE:", mean_absolute_error(y_test,pred_lgb))

print("Ensemble MAE:", mean_absolute_error(y_test,ensemble))
```

```
Random Forest MAE: 0.6069472603683597
XGBoost MAE: 0.6042635805273646
LightGBM MAE: 0.5959314009303006
Ensemble MAE: 0.6019548006189761
```

Results

The results show that all four machine learning models produced accurate predictions. Among them, **LightGBM** achieved the lowest MAE value (**0.5959**), making it the best-performing model for this weather forecasting project.

The **Ensemble Model** also performed well by combining predictions from multiple models. Random Forest and XGBoost produced competitive results, but their prediction errors were slightly higher than LightGBM.

10.Conclusion

This project successfully analyzed global weather data and developed machine learning models to predict temperature values. The dataset was prepared through data preprocessing techniques, including handling missing values, removing duplicate records, removing outliers, and normalizing numerical features.

Exploratory Data Analysis (EDA) was performed to understand weather patterns and relationships between different variables. Anomaly detection using the **Isolation Forest** algorithm helped identify unusual weather observations, while feature importance analysis highlighted the variables that had the greatest impact on temperature prediction.

Several machine learning models were evaluated using the **Mean Absolute Error (MAE)** metric. The comparison showed that **LightGBM** achieved the lowest MAE (**0.5959**), making it the best-performing model among those tested. This demonstrates that advanced machine learning algorithms can provide accurate weather predictions when trained on properly prepared data.

Overall, this project demonstrates a complete data science workflow, including data collection, preprocessing, exploratory analysis, anomaly detection, feature importance analysis, machine learning, and model evaluation. The results show the effectiveness of machine learning techniques in analyzing weather data and generating reliable predictions.

Project Summary

- Cleaned and preprocessed the weather dataset.
- Performed Exploratory Data Analysis (EDA) to identify trends and relationships.
- Detected unusual weather observations using the Isolation Forest algorithm.
- Identified the most influential weather features through feature importance analysis.
- Compared multiple machine learning models using the MAE evaluation metric.
- Determined that **LightGBM** was the best-performing model with the lowest prediction error.